Practical No 08

Aim: Implement Naïve Bayes Classification in Python

```
import pandas as pd

# Dataset
data = {
    'Outlook': [
        'Rainy','Rainy','Overcast','Sunny','Sunny','Sunny',
        'Overcast','Rainy','Rainy','Sunny','Rainy','Overcast',
        'Overcast','Sunny'
    ],
    'Temperature': [
        'Hot','Hot','Hot','Mild','Cool','Cool',
        'Cool','Mild','Cool','Mild','Mild','Mild',
        'Hot','Mild'
    ],
    'Humidity': [
        'High','High','High','High','Normal','Normal',
        'Normal','High','Normal','Normal','Normal','High',
        'Normal','High'
    ],
    'Windy': [
        False, True, False, False, False, True,
        True, False, False, False, True, True,
        False, True
    ],
    'Play Golf': [
        'No','No','Yes','Yes','Yes','No',
        'Yes','No','Yes','Yes','Yes','Yes',
        'Yes','No'
    ]
}

df = pd.DataFrame(data)
print(df)

from collections import Counter

# Separate by class
yes_df = df[df['Play Golf'] == 'Yes']
no_df = df[df['Play Golf'] == 'No']

# Prior probabilities
p_yes = len(yes_df) / len(df)
p_no = len(no_df) / len(df)
print("\nPrior P(Yes) =", p_yes)
print("Prior P(No) =", p_no)

# Prediction query
query = {'Outlook': 'Sunny', 'Humidity': 'High', 'Windy': True}

# Likelihood calculation
def likelihood(df_class, query):
    prob = 1.0
    for feature, value in query.items():
        count = sum(df_class[feature] == value)
        prob *= count / len(df_class) if len(df_class) > 0 else 0
    return prob

likelihood_yes = likelihood(yes_df, query)
likelihood_no = likelihood(no_df, query)

# Posterior using Bayes theorem
posterior_yes = likelihood_yes * p_yes
posterior_no = likelihood_no * p_no

print("\nLikelihood P(X|Yes) =", likelihood_yes)
print("Likelihood P(X|No) =", likelihood_no)
print("Posterior P(Yes|X) =", posterior_yes)
print("Posterior P(No|X) =", posterior_no)

prediction = "Yes" if posterior_yes > posterior_no else "No"
print("\nPrediction: Play Golf =", prediction)
```

```
     Outlook Temperature Humidity  Windy Play Golf
0      Rainy         Hot     High  False        No
1      Rainy         Hot     High   True        No
2   Overcast         Hot     High  False       Yes
3      Sunny        Mild     High  False       Yes
4      Sunny        Cool   Normal  False       Yes
5      Sunny        Cool   Normal   True        No
6   Overcast        Cool   Normal   True       Yes
7      Rainy        Mild     High  False        No
8      Rainy        Cool   Normal  False       Yes
9      Sunny        Mild   Normal  False       Yes
10     Rainy        Mild   Normal   True       Yes
11  Overcast        Mild     High   True       Yes
12  Overcast         Hot   Normal  False       Yes
13     Sunny        Mild     High   True        No

Prior P(Yes) = 0.6428571428571429
Prior P(No) = 0.35714285714285715

Likelihood P(X|Yes) = 0.037037037037037035
Likelihood P(X|No) = 0.19200000000000003
Posterior P(Yes|X) = 0.02380952380952381
Posterior P(No|X) = 0.06857142857142859

Prediction: Play Golf = No
```